Positive Mellin Kernels and Finite-Variance Moment Specifications for the Low-Temperature SK Limiting-Law Claim

Dr. Muralidhara Subbarao
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Stony Brook University
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Abstract

We examine the interfaces between the claimed low-temperature Sherrington–Kirkpatrick limiting law, its finite-tree estimates, and its Mellin specification by finite random experiments. The global spin-glass estimates remain explicit inputs. For the precise logarithmic coefficients used in the source, finite summation telescopes to a positive kernel. We prove discrete normalization and power-bias bounds, an incomplete-gamma size-tail certificate, and an exact account of the resulting large cutoff cost. A source-specific squared-coefficient estimate permits an unbiased single-term experiment with finite variance, a fixed polynomial-tail level law, and replication 8^h instead of the source's 256^h. This does not give finite expected work: even one original size draw has infinite mean. We derive a bounded-frequency characteristic-function certificate and a variance-normalization error bound, keeping finite moments, law identification, and quantitative evaluation distinct. Original local moment and analytic-continuation proofs guide the error bookkeeping; their classical ingredients are credited, not renamed as a new spin-glass proof. Contribution assessment: C3/5, a conditional source-specific finite-variance moment specification and explicit positive-kernel truncation module, without independent certification of the SK limiting law or an efficient sampler.

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1 Model, claim, and the boundary of this report

Fix $\beta > 1$, with zero external field and independent standard real Gaussian disorder g_{ij}. All logarithms are natural. The source normalization is

$$H_n(\sigma) = n^{-1/2} \sum_{1 \le i < j \le n} g_{ij}\sigma_i\sigma_j, \qquad F_n = \log \sum_{\sigma \in \{-1,1\}^n} e^{\beta H_n(\sigma)}. \tag{1}$$

Thus F_n is a total logarithmic partition function, not a free-energy density. Let ζ_n be independent of the disorder, with law $N(0, \beta^2/2)$, and put

$$F_n^c = F_n - n\log 2 + \zeta_n, \quad Y_n = n^{-1/6}(F_n^c - \mathbb{E}F_n^c), \quad X_n = \frac{F_n - \mathbb{E}F_n}{\sqrt{\operatorname{Var} F_n}} \quad (n \ge 2). \tag{2}$$

The completion makes the covariance of the spin field exactly $n\beta^2Q^2/2$; it is not a change of the fluctuation claim. The pinned September 24 manuscript [1] asserts, for each fixed $\beta > 1$, the existence of numbers a_k satisfying

$$|\mathbb{E}Y_n^k - a_k| \le D_k n^{-\alpha_k} \qquad (n \ge 2,\ k \ge 1), \tag{3}$$

$$\sup_{n \ge 2} \mathbb{E}e^{c|Y_n|} \le M, \qquad a_1 = 0, \quad a_2 > 0, \tag{4}$$

where $\alpha_k, D_k > 0$ may depend on k, β, and $c, M > 0$ on β. Changing D_k absorbs finitely many initial sizes. These imply convergence to a moment-determinate law μ_β for Y_n, and a mean-zero, variance-one law for X_n. They also imply $n^{-1/3}\operatorname{Var} F_n \to a_2$.

Logical status. Equations (3)–(4) are *imported source conclusions*, not conclusions established by our finite checks. The local theorems below assume them when applied to SK; as statements about an abstract convergent sequence and its unbiased finite-size observations, their proofs are self-contained. No claim is made that numerical moment agreement, finite-tree identities, or exact symbolic tests independently establish the low-temperature limiting law. Applicability remains L3. The C3 assessment concerns the demonstrated conditional increment, not the truth or importance of the upstream claim.

The physical convention $-F_n/\beta$ reflects the standardized law and multiplies the variance constant by β^{-2}. Dividing additionally by n gives density variance of order $n^{-5/3}$. Neither convention changes a finite-variance estimator into a finite-work algorithm.

2 Source architecture and current context

2.1 The proof chain and its genuine obligations

The source is much more than an ordinary moment argument. Its auxiliary systems have a monotone matrix clock b, field clock h, terminal mass, and positive normalized Gaussian transition kernels. Signed label allocation coefficients are a separate combinatorial device. We retain that separation throughout.

Source component	Role and retained obligation
Foundations and paths	Finite positive tree laws, source Hessians, pinned moment and projection inequalities; a scalar Parisi path and a matrix/field deformation with nonnegative admissible clocks. Simultaneous jumps retain their specified joint filling and intermediate observations.
Pair inverse	An inverse on topology-indexed pair functions. Signed Abel cancellations remove small-mass block denominators before absolute attachment bounds. Uniform domination of resolutions does not assert convergence of the inverse kernels themselves.
Global comparison	Scalar-trial accuracy with a strict power saving, using constrained optimization, common prelimit temperature, coercivity, and a covariance absorption argument. The companion finite-propagation theorem is real input.
Weighted budgets	Auxiliary minimizers, changed density constraints, calibrated profiles, finite strengthening, and buffered one-row closure. A common envelope, not an arbitrary measurable choice of optimizer, controls parameter averages.
Statistics	Deterministic windows, residual blocks, higher pinned moments, overlap alias replacement, and fixed-marginal centered integration. Ordinary unconditional Hölder is used only after constructing the retained law.
Pressure expansion	Finite Taylor expansions and low-degree signed cancellations. The size operator is $1+\partial_\theta$, with the *formal* assignment $\partial_\theta\Delta = -\Delta/6$, not differentiation of a random overlap.
Reduction and parameters	Stopped Laplace comparisons on all integer dyads; finite interpolation extracts cumulants, summable comparisons give moment limits, and a separate lower-scale input gives $a_2 > 0$. Exponential integrability then identifies the law. Parameter orders precede $n \to \infty$.
Mellin specification	Expectations of size-randomized finite experiments are summed via analytic continuation. The pointwise coefficient series is zero and cannot replace this order of operations. This is the principal interface refined here.

In the reduction, $x = N^{-1/6}$ and a small positive Laplace test has scale x^ε. The dyadic comparison has a saving independent of the subsequently chosen sufficiently small ε. Extracting a fixed kth coefficient loses powers of $x^{-\varepsilon}$, so one fixes k and then chooses ε within the available saving. This is not a simultaneous error estimate for all moments. The source's positivity proof for a_2 uses a power-rate upper bound under a hypothetical zero limit against a companion subpower lower bound. Mere variance-exponent notation would not suffice.

The source correctly sums its signed scalar coefficients before taking absolute values. In particular, the degree-two square counterterm must be combined with the repeated-pair coefficient on the same

projected skeleton. Replacing the signed allocation rule by a probability rule, or introducing a common tilted root for genuinely separate components, would invalidate this mechanism.

2.2 Public literature checked on October 9, 2026

Aronow and Lopatto's September 2026 preprint [2] uses the same logarithmic partition-function convention. Its Theorem 1.4 states, for fixed $\beta > 1$, variance bounds between constant multiples of $n^{4/15}$ and $n^{7/15}$ for sufficiently large n. We checked that statement and normalization, not its 112-page proof. It neither supplies the exact $n^{1/3}$ variance asymptotic nor identifies the present limiting law.

Lopatto's updated October 5 version [3] states interval support, smooth interior density, and a terminal atom for the zero-field Parisi measure at every $\beta > 1$. The source cites the earlier v3; our update is an abstract/version check, not a new verification of its support theorem. Du and Huang [4] state a critical-temperature Gaussian limit and variance $\frac{1}{6}\log n + O(1)$ at $\beta = 1$. That different regime must not be substituted for fixed $\beta > 1$.

Single-term randomization and its variance/work tradeoffs are classical unbiased-estimation ideas, notably Rhee–Glynn [5]. Our contribution is not that general idea. It is the positive telescoping structure and coefficient growth control for this source's particular Mellin experiment, together with a fully specified level law and cutoff budget. No independent historical priority claim is made for the resulting elementary kernel identities.

3 The precise finite experiment

Fix a moment order $k \geq 1$. Write $m_\ell = \mathbb{E}Y_\ell^k$ and $a = a_k$. For a chosen size ℓ, take $k+1$ independent copies of F_ℓ^c and define

$$P_\ell = \prod_{i=1}^{k} \ell^{-1/6}(F_\ell^{c,0} - F_\ell^{c,i}). \tag{5}$$

The differences share copy 0 and are not independent. Conditioning on that copy nevertheless gives exactly

$$\mathbb{E}[P_\ell \mid F_\ell^{c,0}] = \{\ell^{-1/6}(F_\ell^{c,0} - \mathbb{E}F_\ell^c)\}^k, \qquad \mathbb{E}P_\ell = m_\ell. \tag{6}$$

Uniform $2k$th moments of Y_ℓ imply $\sup_\ell \mathbb{E}P_\ell^2 =: B < \infty$: Hölder bounds the product of k squared differences by the product of their $2k$-norms squared, and $\|Y_\ell^0 - Y_\ell^i\|_{2k} \leq 2\|Y_\ell\|_{2k}$. This is an existence bound depending on k, β.

Let

$$C_0 = \sum_{\ell=2}^{\infty} \ell^{-2}, \qquad \mathbb{P}(J = \ell) = \ell^{-2}/C_0, \tag{7}$$

and, conditionally on $J = \ell$, use the experiment above. For $h \geq 0$ set

$$d_h(t) = \frac{t^h}{h!} - \mathbf{1}_{\{h\geq 1\}}\frac{t^{h-1}}{(h-1)!}, \qquad A_h = C_0 P_J d_h(\log J), \quad \mu_h = \mathbb{E}A_h. \tag{8}$$

For every fixed h these variables have finite second moments. The abstract results below need only $\mathbb{E}P_\ell = m_\ell$, $\sup_\ell \mathbb{E}P_\ell^2 \leq B$, and

$$|m_\ell - a| \leq D\ell^{-\alpha} \quad (\ell \geq 2), \qquad D < \infty, \quad \alpha > 0. \tag{9}$$

They do not require independence between the k factors in (5); different *replications* will be independent.

3.1 Why expectation must precede the infinite sum

For $\Re z > 0$ define

$$\mathcal{B}(z) = z \sum_{\ell=2}^{\infty} m_\ell \ell^{-1-z}. \tag{10}$$

The source proves continuation across $z = 0$ and $\mathcal{B}(0) = a$. Here is the elementary domain argument. The series

$$D_0(z) = \sum_{\ell=2}^{\infty} \ell^{-1-z} - \int_1^{\infty} u^{-1-z}\,\mathrm{d}u$$

initially equals a sum of cell discrepancies. Each cell difference is bounded on compact subsets of $\Re z > -1$ by a constant times $\ell^{-2-\Re z}$, using the derivative of u^{-1-z}. Thus the discrepancy, continued by those cells, is holomorphic there. Writing $m_\ell = a + e_\ell$ gives

$$\mathcal{B}(z) = a\{1 + zD_0(z)\} + z \sum_{\ell=2}^{\infty} e_\ell \ell^{-1-z}. \tag{11}$$

The last series is locally uniformly convergent on $\Re z > -\alpha$. Hence $\mathcal{B}$ is holomorphic on $\Re z > -\min(1, \alpha)$, including 0, with value a.

For u near zero, $\mathcal{B}(1-u) = (1-u)\sum m_\ell \ell^{-2} e^{u \log \ell}$. Termwise differentiation is justified on a smaller disk by a summable $\ell^{-2+\delta}(1 + \log^h \ell)$ majorant. Its u^h coefficient is μ_h. Choosing any

$$1 < \rho < 1 + \min(1, \alpha), \qquad C_\rho = \max_{|u|=\rho} |\mathcal{B}(1-u)|, \tag{12}$$

Cauchy's formula gives $|\mu_h| \le C_\rho \rho^{-h}$ and

$$\sum_{h=0}^{\infty} |\mu_h| < \infty, \qquad \sum_{h=0}^{\infty} \mu_h = a. \tag{13}$$

In contrast, for each finite t, $\sum_{h \ge 0} d_h(t) = e^t - e^t = 0$. Consequently $\sum_h A_h = 0$ almost surely when all coefficients use one realization of (J, P_J). It is invalid to take its expectation as a substitute for (13). This distinction is already explicit in the source; we retain it rather than presenting it as a source correction.

4 Positive finite sums and quantitative truncation

Lemma 4.1 (The exact finite telescoping kernel)**.** *For every integer $H \ge 0$ and every real t,*

$$\sum_{h=0}^{H} d_h(t) = t^H / H!. \qquad S_H := \sum_{h=0}^{H} \mu_h = \sum_{\ell=2}^{\infty} m_\ell\, w_H(\ell), \quad w_H(u) = \frac{u^{-2}(\log u)^H}{H!}.$$

The weights are nonnegative for $u \ge 1$.

Proof. Adjacent factorial terms cancel in the finite sum. Its expectation is legitimate because there are finitely many integrable terms. Substitution of the prior in (7) gives the displayed series. □

Positive weights remove coefficient cancellation in this finite *expectation* approximation. They do not make P_ℓ nonnegative or convert the estimator into a sample from the limiting law.

Lemma 4.2 (Discrete logarithmic-moment budget). *For $\lambda > 0$ and integer $H \geq 1$, put*

$$Q_H(\lambda) = \sum_{\ell=2}^{\infty} \frac{\ell^{-1-\lambda}(\log \ell)^H}{H!}.$$

Then

$$\left| Q_H(\lambda) - \lambda^{-H-1} \right| \leq 2(1+\lambda)^{-H}. \tag{14}$$

In particular, with $Z_H = Q_H(1)$, $|Z_H - 1| \leq 2^{1-H}$ and $Z_H \geq 3/4$ for $H \geq 3$. Also $Z_0 = C_0 < 1$.

Proof. For $f(u) = u^{-1-\lambda}(\log u)^H/H!$, substitution $t = \log u$ gives $\int_1^\infty f = \lambda^{-H-1}$. On the cells $[\ell - 1, \ell]$ the sum-integral error is at most $\int_1^\infty |f'(u)|\,\mathrm{d}u$. Since $f(1) = 0$, $f(\infty) = 0$, and f has exactly one maximum, this variation equals $2 \max f$. The maximum occurs at $\log u = H/(1+\lambda)$ and is

$$\max f = \frac{(H/((1+\lambda)e))^H}{H!} \leq (1+\lambda)^{-H}.$$

The last inequality follows, for example, from $e^H = \sum_{j \geq 0} H^j/j! \geq H^H/H!$. For $H = 0, \lambda = 1$, monotonicity gives $C_0 = \sum_{\ell \geq 2} \ell^{-2} < \int_1^\infty u^{-2}\,\mathrm{d}u = 1$. □

Theorem 4.3 (Normalized positive approximation). *Under (9), for every $H \geq 3$,*

$$\left| \frac{S_H}{Z_H} - a \right| \leq \frac{D}{Z_H} \left\{ (1+\alpha)^{-H-1} + 2(2+\alpha)^{-H} \right\}. \tag{15}$$

For the unnormalized partial sum, add $|a|\,2^{1-H}$ to $D\{(1+\alpha)^{-H-1} + 2(2+\alpha)^{-H}\}$.

Proof. The normalized weights sum to one, so subtraction of a leaves only $m_\ell - a$. Positivity and (9) bound its absolute weighted sum by $DQ_H(1+\alpha)$. Apply Lemma 4.2. The unnormalized formula additionally contains $a(Z_H - 1)$. □

This proof identifies an explicit bias without evaluating a complex continuation or estimating C_ρ. It still needs a quantitative finite-size moment bound D, α. Existence of unspecified constants is not a computable numerical tolerance.

Proposition 4.4 (A certified finite size cutoff). *Let $H \geq 3$, $L \geq 2$ an integer, and $\log L \geq H/2$. Set*

$$Z_{H,L} = \sum_{\ell=2}^{L} w_H(\ell), \quad S_{H,L} = \sum_{\ell=2}^{L} m_\ell w_H(\ell), \quad U_H(L) = L^{-1} \sum_{j=0}^{H} \frac{(\log L)^j}{j!}.$$

Then $0 \leq Z_H - Z_{H,L} \leq U_H(L)$ and

$$\left| \frac{S_{H,L}}{Z_{H,L}} - a \right| \leq \frac{D\{(1+\alpha)^{-H-1} + 2(2+\alpha)^{-H}\}}{1 - 2^{1-H} - U_H(L)} \tag{16}$$

whenever the displayed denominator is positive. If measured moments $\widehat{m}_\ell$ have certified errors η_ℓ, add $Z_{H,L}^{-1} \sum_{\ell=2}^{L} w_H(\ell)\eta_\ell$.

Proof. The weight w_H decreases for $u \geq e^{H/2}$. The decreasing right sum above L is at most $\int_L^\infty w_H(u)\,\mathrm{d}u$. Substitution and H integrations by parts evaluate that integral as $U_H(L)$. Therefore $Z_{H,L} \geq 1 - 2^{1-H} - U_H(L)$. In the numerator after subtracting a, positivity bounds the truncated error by the full $DQ_H(1+\alpha)$. The measured-moment error is a separate finite weighted sum by the triangle inequality. □

Corollary 4.5 (An explicit cost scale)**.** *If* $\log L \geq c_*(H+1)$ *with* $c_* > 1$*, then*

$$U_H(L) \leq \exp\{-(H+1)(c_* - 1 - \log c_*)\}. \tag{17}$$

Proof. The integral defining $U_H(L)$ is the upper tail of a gamma random variable of integer shape $H+1$ and rate one. Its exponential moment at $s \in (0,1)$ is $(1-s)^{-H-1}$, obtained by integration. Markov's inequality with $s = 1 - c_*^{-1}$ proves the bound. □

The logarithmic-size distribution concentrates near $H+1$, not near a fixed size. Increasing H suppresses bias by moving weight to large systems. A literal summation through L grows exponentially in H; even one SK partition function at size ℓ is not assumed cheaply available. No claimed complexity improvement follows from positivity.

Example 4.6 (An exact error certificate with an expensive cutoff)**.** Consider any manufactured sequence with $|m_\ell - a| \leq \ell^{-1}$. Take $H = 12$ and $L = 3^{26} = 2541865828329$. Since $e < 3$, $\log L > 26 = 2(H+1)$. Also $e > 8/3$, so (17) is less than $(3/4)^{13}$. Thus the bias is at most the exact rational number

$$\frac{2^{-13} + 2\,3^{-12}}{1 - 2^{-11} - (3/4)^{13}} < \frac{1}{7500}.$$

This is not an evaluated SK moment or a recommended computation. It exhibits both a usable conditional bound and its prohibitive literal size cost.

5 Squared coefficients and finite-variance replication

5.1 The source-specific energy identity

Theorem 5.1 (Discrete coefficient energy)**.** *For* $h \geq 1$*, let*

$$c_h = \binom{2h-2}{h-1}, \qquad j_0 = \tfrac{1}{2}, \qquad j_h = \frac{c_h}{h\,4^h}, \qquad \epsilon_h = 2\{j_h + \sqrt{j_h j_{h-1}}\}.$$

Then

$$\int_0^\infty e^{-t} d_h(t)^2 \,\mathrm{d}t = c_h, \tag{18}$$

$$\int_0^\infty e^{-2t} d_h(t)^2 \,\mathrm{d}t = j_h, \tag{19}$$

$$\left| \sum_{\ell=2}^{\infty} \ell^{-2} d_h(\log \ell)^2 - c_h \right| \leq \epsilon_h \leq \tfrac{3}{2}. \tag{20}$$

In particular, for all $h \geq 0$*,*

$$\mathbb{E} A_h^2 \leq C_0 B\, 4^h. \tag{21}$$

Proof. The elementary gamma integral $\int_0^\infty e^{-\lambda t} t^r \,\mathrm{d}t = r!\lambda^{-r-1}$ follows by integration by parts. Expanding the square gives, for $h \geq 1$,

$$I_h(\lambda) = \frac{(2h)!}{h!^2 \lambda^{2h+1}} - \frac{2(2h-1)!}{h!(h-1)!\lambda^{2h}} + \frac{(2h-2)!}{(h-1)!^2 \lambda^{2h-1}}. \tag{22}$$

At $\lambda = 1$ the first two terms cancel exactly. At $\lambda = 2$, using $\binom{2h}{h} = (4 - 2/h)c_h$, their combination with the last term is $c_h/(h4^h)$. This proves the two integral identities.

For $f_h(u) = u^{-2} d_h(\log u)^2$, the same cell comparison as before gives a sum-integral discrepancy at most $\int_1^\infty |f_h'(u)|\,\mathrm{d}u$. Since $d_h' = d_{h-1}$ for $h \geq 1$, the substitution $t = \log u$ yields

$$\int_1^\infty |f_h'| \leq 2 \int_0^\infty e^{-2t} \{d_h(t)^2 + |d_h(t) d_{h-1}(t)|\}\,\mathrm{d}t \leq 2(j_h + \sqrt{j_h j_{h-1}}).$$

Here $I_0(2) = 1/2$. The central binomial coefficient satisfies $c_h \leq 4^{h-1}$, so $j_h \leq 1/(4h)$. For $h = 1$, $\epsilon_1 = (1+\sqrt{2})/2 < 3/2$; for $h \geq 2$, $j_h, j_{h-1} \leq 1/4$ gives $\epsilon_h \leq 1$. This proves (20).

The prior and conditional second-moment bound give

$$\mathbb{E}A_h^2 \leq C_0 B \sum_{\ell \geq 2} \ell^{-2} d_h(\log \ell)^2 \leq C_0 B(4^{h-1} + 3/2) \leq C_0 B 4^h \quad (h \geq 1).$$

For $h = 0$, $\mathbb{E}A_0^2 \leq C_0^2 B < C_0 B$. Thus the bound includes the initial coefficient. □

The classical integral and total-variation estimates are elementary. Their useful source-specific feature is the exact cancellation in (18): the L^2 growth is at most 2^h in norm, whereas the source's sufficient estimate grows like 4^h in norm. We make no assertion that the source's looser bound is incorrect.

5.2 An unbiased specification with finite variance

Theorem 5.2 (Fixed-prior, finite-variance specification)**.** *Assume* (9), $\mathbb{E}P_\ell = m_\ell$, *and* $\sup_\ell \mathbb{E}P_\ell^2 \leq B$. *Independently choose an integer level* $H \geq 0$ *with*

$$p_h = \mathbb{P}(H = h) = \frac{1}{(h+1)(h+2)}. \tag{23}$$

Conditionally on $H = h$, *generate* $R_h = 8^h$ *independent copies of* A_h, *each with its own size draw and its own finite experiment. Let* $\overline{A}_h$ *be their average and set*

$$W = \overline{A}_H / p_H. \tag{24}$$

Then $W \in L^2$ *and* $\mathbb{E}W = a$. *More explicitly, for any* ρ, C_ρ *as in* (12),

$$\mathbb{E}W^2 \leq 16 C_0 B + \frac{2C_\rho^2}{(1-\rho^{-2})^3}. \tag{25}$$

All level and replication choices are independent of the unknown positive convergence exponent α.

Proof. The identity $p_h = (h+1)^{-1} - (h+2)^{-1}$ shows that the level law has total mass one. Conditional independence of the replications gives

$$\mathbb{E}W^2 = \sum_{h=0}^\infty \frac{\mathrm{Var}(A_h)/R_h + \mu_h^2}{p_h}.$$

This equality follows by Tonelli applied to the nonnegative square, without presuming its finiteness. By Theorem 5.1, its first part is bounded by $C_0 B \sum (h+1)(h+2) 2^{-h} = 16 C_0 B$. For its second part use $|\mu_h| \leq C_\rho \rho^{-h}$ and

$$\sum_{h=0}^\infty (h+1)(h+2) z^h = \frac{2}{(1-z)^3} \quad (0 \leq z < 1),$$

obtained by differentiating the geometric series twice. This proves finiteness and (25). In particular W is integrable, and conditioning on H yields $\mathbb{E}W = \sum_h \mu_h = a$ by (13). □

The source uses a geometric level law 2^{-h-1} and replication 256^h to obtain an integrable unbiased variable. Merely reducing its replication does not by itself prove finite variance: the deterministic term $\sum_h \mu_h^2/p_h$ must be controlled as well. A geometric level law can fail that test when the only available analytic radius is close to one. The polynomial-tail law in (23) works for every $\rho > 1$, while the sharper energy estimate controls its random part. This two-part check is the reason for changing the level law as well as the replication count.

Proposition 5.3 (The geometric replication threshold in a test family)**.** *For the prior* (23)*, any integer geometric replication $R_h = b^h$ with integer $b > 4$ gives the same conclusion as Theorem 5.2. The sufficient geometric threshold* 4 *cannot be reduced over all experiments satisfying its assumptions: in the manufactured example $P_\ell = 1$, $m_\ell = a = 1$, every integer $1 \le b \le 4$ gives $\mathbb{E}W^2 = \infty$.*

Proof. For $b > 4$ replace 2^{-h} in the noise sum by $(4/b)^h$. For the counterexample, the variance is exactly

$$\operatorname{Var} A_h = C_0 \sum_{\ell \ge 2} \ell^{-2} d_h (\log \ell)^2 - \mu_h^2.$$

Here μ_h decays geometrically by (11) for every $\rho < 2$. Equation (20) therefore makes $\operatorname{Var} A_h \ge C_0 c_h/2$ for sufficiently large h. The central coefficient is the largest coefficient of $(1+z)^{2h-2}$, so $c_h \ge 4^{h-1}/(2h-1)$. Consequently the noise summand $\operatorname{Var}(A_h)/(p_h b^h)$ is at least

$$\frac{C_0}{8} \frac{(h+1)(h+2)}{2h-1} \left(\frac{4}{b}\right)^h$$

eventually. For $b \le 4$ it does not even tend to zero. □

This is a sharp geometric-growth boundary for the stated general class and level law, not an optimality theorem for SK computation. Polynomial factors in a base-4 replication, different importance laws, or coupled experiments are distinct design choices.

5.3 What is, and is not, finite

The level H, each replication count, and each sampled size J are finite almost surely. A finite SK system has a finite partition sum, so this is an almost-surely finite mathematical experiment, under the same ideal real-randomness convention as the source. Exact real Gaussian simulation or exact transcendental arithmetic is not a finite-bit implementation claim. Moreover,

$$\mathbb{E}R_H = \sum_{h \ge 0} \frac{8^h}{(h+1)(h+2)} = \infty, \qquad \mathbb{E}J = C_0^{-1} \sum_{\ell \ge 2} \ell^{-1} = \infty. \tag{26}$$

Thus even a cost model charging one unit per replica, or only ℓ units per size-ℓ system, has infinite expected work. Finite variance permits an ordinary iid sample-mean variance statement at a *fixed number of completed experiments*. It does not permit a square-root wall-clock claim or a finite expected runtime bound.

The normalization C_0 and probabilities in (7) are exact mathematical objects. Rounding their values requires an additional sampling/arithmetic error analysis. Likewise, $\mathbb{E}W = a_k$ is a specification of a moment, not a way to draw a random variable with law μ_β; independent moment estimators need not form a realizable moment table.

6 From moment certificates to characteristic functions

Theorem 6.1 (A bounded-frequency error budget)**.** *Let Y have the limiting law μ_β, and assume $\mathbb{E}e^{c|Y|} \le M$. For $K \ge 1$ suppose deterministic estimates $\widehat{a}_j$ obey $|\widehat{a}_j - \mathbb{E}Y^j| \le \eta_j$ for $1 \le j \le K$, and set $\widehat{a}_0 = 1$. Then for every real t,*

$$\left| \mathbb{E}e^{itY} - \sum_{j=0}^{K} \frac{(it)^j}{j!} \widehat{a}_j \right| \le \sum_{j=1}^{K} \frac{|t|^j}{j!} \eta_j + M \left(\frac{|t|}{c} \right)^{K+1}. \tag{27}$$

The last term decays with K only for $|t| < c$. For comparing Y_n to Y with the same exponential bound, one may instead use the moment discrepancies from (3) *and twice the displayed remainder.*

Proof. Taylor's integral remainder for the real parameter on $u \mapsto e^{itu}$ has magnitude at most $|tY|^{K+1}/(K+1)!$, since $|e^{itv}| = 1$ for real v. The exponential moment bounds $\mathbb{E}|Y|^{K+1} \le M(K+1)!c^{-K-1}$. Take expectations of the finite formula, then insert the certified moment errors. Apply the same argument twice for a comparison of laws. □

For the SK limit, the bound on Y follows from the uniform source bound by weak convergence and truncation, or the lower-semicontinuous version of Fatou's lemma. It is not supplied by a finite observed moment table. Proposition 4.4 can supply the individual η_j only when the corresponding D_j, α_j and finite-data errors are actually certified. An unbiased sample mean requires a specified probability confidence statement in place of deterministic η_j.

Corollary 6.2 (Variance-normalization sensitivity)**.** *Suppose $a_2 > 0$, $\mathbb{E}Y = 0$, and a deterministic $\widehat{v}$ satisfies $|a_2 - \widehat{v}| \le \eta_2 < \widehat{v}$. Define $u = t/\sqrt{\widehat{v}}$. The error in replacing the characteristic function of $Y/\sqrt{a_2}$ by the polynomial in* (27) *evaluated at u is at most its right side at u, plus*

$$\frac{|t|\eta_2}{\sqrt{\widehat{v}}\{\sqrt{\widehat{v} - \eta_2} + \sqrt{\widehat{v}}\}}. \tag{28}$$

Proof. The inequality $|e^{ix} - e^{iy}| \le |x - y|$ and $\mathbb{E}|Y| \le \sqrt{a_2}$ bound the scale-change error by $|t||\sqrt{a_2/\widehat{v}} - 1|$. Rationalize the square-root difference and use $a_2 \ge \widehat{v} - \eta_2$. □

The positive denominator is essential. Neither a nearly zero empirical variance nor analytic continuation alone controls standardization. The finite polynomial in (27) also need not be a characteristic function: positive definiteness and global modulus at most one are additional law constraints, not consequences of a local error budget.

Removing the independent completion. Writing $\widetilde{Y}_n = n^{-1/6}(F_n - \mathbb{E}F_n)$, independence gives exactly

$$\varphi_{Y_n}(t) = \varphi_{\widetilde{Y}_n}(t) \exp\{-\beta^2 t^2/(4n^{1/3})\}. \tag{29}$$

Therefore removing the completion multiplies a characteristic-function error at frequency t by at most $\exp\{\beta^2 t^2/(4n^{1/3})\}$. For a direct comparison, $|\varphi_{Y_n}(t) - \varphi_{\widetilde{Y}_n}(t)|$ is at most $1 - \exp\{-\beta^2 t^2/(4n^{1/3})\}$. This controlled bounded-frequency operation is not a justification for unstable large-frequency deconvolution.

7 Finite marginals, private originals, and error types

7.1 A retained law is not a conditioned law

The source's centered path integral must be evaluated under a law which does not change when its integration variable moves. After overlap aliases are replaced, unused branches can be pruned because their continuation kernels have mass one. An old pair's fixed split and fixed projection cuts remain in the retained law. Conditioning on a terminal observation on the branch being pruned is a different operation.

Example 7.1 (A two-branch conditioning obstruction)**.** Let U be uniform on $\{-1, 1\}$ and let $V = U\epsilon$, where ϵ is independent of U, takes values ± 1, and $\mathbb{E}\epsilon = \rho \in (0, 1)$. Marginalizing the normalized transition from U to V leaves $\mathbb{E}U = 0$. Conditioning instead on $V = 1$ gives $\mathbb{E}[U \mid V = 1] = \rho$. Thus even the simplest normalized branch cannot be replaced by its observed conditional law without changing an earlier observable.

A useful finite implementation check is a kernel-error ledger. If finitely many positive probability kernels K_i are replaced by positive probability kernels $\widehat{K}_i$ with $\sup_x \|K_i(x, \cdot) - \widehat{K}_i(x, \cdot)\|_{\mathrm{TV}} \le e_i$, then any bounded retained observable f changes in expectation by at most $2\|f\|_\infty \sum_i e_i$, with TV defined as half the total variation norm. Replace kernels one at a time; the surrounding probability kernels are contractions and the middle signed measure has norm at most $2e_i$. For a signed topology sum, multiply the per-topology error by its absolute coefficient and sum. This standard argument does not prove a uniform continuum estimate unless those absolute sums and discarded-tree errors are controlled. It is included as an interface safeguard, not a separate notable contribution.

7.2 Fresh original proof support

The original local-moment chapter [6] proves the finite Taylor integral remainder by integrating the one-dimensional segment formula against a kernel with the required absolute moment. It separately distinguishes output normalization from total-mass normalization, polynomial exactness from closure, and local error from discarded tails. The uses here are concrete: (27) integrates a finite Taylor remainder under an exponential-moment majorant, and (16) retains the normalization lost through truncation. Neither imports a finite inverse theorem or a physical imaging model.

The original analytic-continuation proof [7] requires a common holomorphic domain and locally uniform integrable majorants. Its identity theorem is explicitly uniqueness, not existence or stability. In (11) we therefore construct the actual common domain and convergent cell-discrepancy series instead of invoking a general Ram continuation principle. Finite noisy samples of $\mathcal{B}$ do not determine a stable evaluation at zero without additional bounds.

The later core report [8] was freshly read as a comparison, not substituted for those originals. Its covariance calculation needs a common joint law and sufficiently high moments; its remainder ledger keeps approximation error separate from random variance. This is exactly why the shared-copy factors in (5) are conditioned before their expectation is taken, and why the deterministic mean term and replication variance are both retained in (25).

8 Advantages, costs, and feasible extensions

What the local module improves. Positive finite sums avoid alternating coefficient cancellation when approximating a moment. The sum-integral variation certificate gives explicit constants, and the size cutoff has an elementary finite incomplete-gamma formula. The sharper squared-coefficient

control permits a finite-variance unbiased specification with a level prior that does not depend on an unknown convergence exponent. The test family in Proposition 5.3 identifies a genuine geometric replication obstruction, rather than merely a sufficient parameter choice. These are conditional, source-specific refinements of a classical method.

Costs and limitations. The limiting moments, rates, exponential bound, and positive variance still depend on the source's global hierarchy and companion inputs. Their constants have not been made numerically effective here. The positive kernel drifts to exponentially large sizes; the unbiased experiment has infinite expected replication and size costs. Finite variance is not low variance, nor does it bound runtime tails. The present proof neither identifies the limiting density in closed form nor shows it to be Gaussian, nor establishes uniformity as $\beta \downarrow 1$. It gives no new result for external field, growing moment order, or a temperature depending on n.

Extensions with explicit obligations. The positive-kernel argument works for signed or complex m_ℓ under the same absolute rate bound. Several moment orders can share a finite size cutoff, provided all required rates and errors are retained separately. Coupling coefficient replications across levels or changing the size importance law could change the work/variance tradeoff; doing so requires a new unbiasedness and second-moment calculation, not just reusing (25). A finite expected-work design would need to address both the level distribution and the heavy size law.

For practical characteristic-function evaluation, a worthwhile next step is certified arithmetic for the finite logarithmic weights and the actual finite-system moments. Another is an effective version of the source's power rates and exponential-moment constants. Neither is supplied by a finite rank computation. A genuinely quantitative law metric would also require a justified smoothing inequality and large-frequency or regularity information; our small-frequency polynomial should not be advertised as such a metric.

9 Read scope, checks, and assessment

The accompanying read ledger records literal bounded displays and pre/post SHA-256 hashes. All source sections except the large middle portion of the weighted-budget proof were read in full, including the scientific proofs of the reduction, inverse, global comparison, statistics, and pressure expansion. In `07-budgets.tex`, lines 1–236 and 1850–2253 were read; lines 237–1849 remain unread and imported. Bibliography, figures, consequences, and notation were read separately. Reading these files does not amount to independently auditing every imported theorem or the companion manuscript.

Fresh private literal scope is the local-moment chapter, lines 1–446; TR018, lines 1155–1244; and core16, lines 216–374. Included child files of the chapter and the remaining corpus are not claimed read. The catalogue's TR6 locator was prehashed but not read or used as a proof input. Public literature statement/metadata checks are recorded separately from manuscript reads. Parent subtree authentication is snapshot provenance only; parent checks are not counted as author tests.

The owned exact-check script tests the logarithmic polynomial telescoping, gamma integrals, coefficient-energy cancellation, geometric generating function, finite shared-copy moment identity, conditional-branch example, and rational cutoff and replication bounds. These are finite manufactured algebraic fixtures, not SK simulations or global proof certificates. The run passed 533 cases and 1,862 exact assertions, recorded separately in the test receipt. No foreign checker was executed. The proof of each infinite assertion is in the text rather than inferred from these fixtures.

No mathematical correction to the source is asserted on the basis of this scope. Its expectation-before-series order, finite-marginal normalization, signed cancellation order, and distinction between almost-sure finiteness and expected work are retained. The only updated context is the explicitly versioned public literature. The assessment is C3 because Theorems 4.3, 5.1, and 5.2, with Propositions 4.4 and 5.3, form a usable conditional refinement of the actual source construction. It is not C4: none of the central spin-glass inputs has been replaced or independently established, and no efficient computation of the limiting law is proved.

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